



**RIVER VALLEY HIGH SCHOOL
H2 CHEMISTRY**

Paper 1

<i>Question Number</i>	<i>Key</i>	<i>Question Number</i>	<i>Key</i>
1	C	16	B
2	B	17	A
3	D	18	D
4	C	19	A
5	D	20	C
6	B	21	A
7	B	22	A
8	B	23	D
9	C	24	C
10	B	25	D
11	A	26	A
12	A	27	A
13	D	28	B
14	D	29	C
15	A	30	A

Suggested solutions for Paper 2

- 1 (a) (i) $A^+(g) \rightarrow A^{2+}(g) + e^-$ ΔH : 2nd IE of A
 (Note: state symbols need to be shown)
- (ii) The successive ionisation energies increase generally as electrons are being removed from an ion of increasing positive charge. Hence the electron to be removed experiences an increasingly stronger electrostatic attraction, requiring more energy to remove.
- The 7th electron is the last electron to be removed from the outermost principal quantum shell and the 8th electron is removed from an inner principal quantum shell which is located nearer the nucleus. Hence the 8th electron experiences a greater effective nuclear charge and is more strongly attracted to the nucleus. A much larger amount of energy is required to remove the 8th electron.
- (iii) Chlorine
 $2s^2 2p^6 3s^2 3p^5$
- (b) (i) Hydrogen has one valence electron, just like the Group 1 elements. It will be at the top of the group as only one principal quantum shell is occupied.
- (ii) Unlike Group 1 elements which have giant metallic structure consisting of cations surrounded by the sea of delocalised valence electrons, hydrogen has simple covalent structure consisting of discrete molecules.
- Hydrogen gas has a lower melting/boiling point than the Group 1 metals / hydrogen gas does not conduct electricity in solid state unlike Group 1 metals.

2 (a) $[\text{HCO}_2\text{H}]_{\text{initial}} = 0.0100 \div \frac{250}{1000} = 0.0400 \text{ mol dm}^{-3}$

	HCO_2H	$\rightleftharpoons$	HCO_2^-	+	H^+
Initial [] / mol dm^{-3}	0.0400		0		0
Change in [] / mol dm^{-3}	-x		+x		+x
Equilibrium [] / mol dm^{-3}	$0.0400 - x$		x		x

$$K_a = \frac{(x)(x)}{0.0400 - x}$$

Assuming x is small such that $0.0400 - x \approx 0.0400$

$$1.60 \times 10^{-4} = \frac{x^2}{0.0400}$$

$$x = 2.53 \times 10^{-3}$$

$$\text{pH} = -\lg(2.53 \times 10^{-3}) = \underline{2.60}$$

(b) Let $[\text{HCO}_2\text{Na}]$ in solution **B** be $y \text{ mol dm}^{-3}$.

$$\text{No. of moles of } \text{HCO}_2\text{Na} = \frac{100}{1000} \times y = 0.1y \text{ mol}$$

$$[\text{HCO}_2\text{Na}] \text{ in buffer solution} = 0.1y \div \frac{250}{1000} = 0.4y \text{ mol dm}^{-3}$$

$$\text{No. of moles of } \text{HCO}_2\text{H} = \frac{150}{1000} \times 0.0100 = 0.0150 \text{ mol}$$

$$[\text{HCO}_2\text{H}] \text{ in buffer solution} = 0.0150 \div \frac{250}{1000} = 0.00600 \text{ mol dm}^{-3}$$

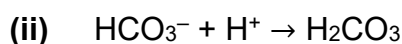
$$\text{pH} = \text{p}K_a + \lg \frac{[\text{salt}]}{[\text{acid}]}$$

$$3.70 = -\lg(1.60 \times 10^{-4}) + \lg \frac{0.4y}{0.00600}$$

$$y = 0.0120$$

$$[\text{HCO}_2\text{Na}] = \underline{0.0120 \text{ mol dm}^{-3}}$$

(c) (i) pH of blood will decrease.



The large amount of HCO_3^- reacts with the small amount of H^+ from the lactic acid to form H_2CO_3 , which dissociates to give CO_2 and H_2O . This prevents the $[\text{H}^+]$ from increasing appreciably and allows the pH to stay approximately constant.

3 (a) (i)

	C	H	O
Mass ratio	54.5	9.10	36.4
Mole ratio	$\frac{54.5}{12.0} = 4.54$	$\frac{9.10}{1.0} = 9.10$	$\frac{36.4}{16.0} = 2.275$
Simplest ratio	$\frac{4.54}{2.275} = 2.00 = 2$	$\frac{9.10}{2.275} = 4.00 = 4$	$\frac{2.275}{2.275} = 1.00 = 1$

Empirical formula is $\text{C}_2\text{H}_4\text{O}$

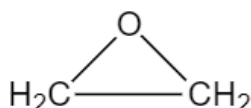
Let molecular formula be $(\text{C}_2\text{H}_4\text{O})_n$

$$(12.0 \times 2 + 1.0 \times 4 + 16.0)n = 44$$

$$n = 1$$

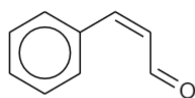
Hence, molecular formula is $\text{C}_2\text{H}_4\text{O}$.

(ii)



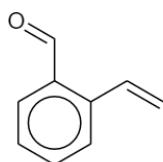
- (b) aqueous bromine – orange $\text{Br}_2(\text{aq})$ decolourises (Note: no white ppt is formed)
 aqueous sodium carbonate – no effervescence observed and no gas evolved
- (c) A very small K_c value that is less than 1 suggests that the equilibrium position lies very far to the left.
 The major species is ethanal.
- (d) $\text{CH}_3\text{CHO}(\text{l}) + 3\text{I}_2(\text{aq}) + 4\text{OH}^-(\text{aq}) \rightarrow \text{CHI}_3(\text{s}) + \text{HCOO}^-(\text{aq}) + 3\text{I}^-(\text{aq}) + 3\text{H}_2\text{O}(\text{l})$
 (Note: common incorrect answers included the incorrect carboxylate ion or CH_3I as the iodoform product)

4 (a) isomer E

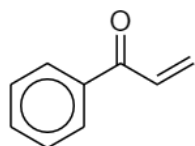


cis-trans isomer

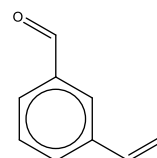
isomer F



or



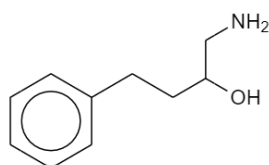
or



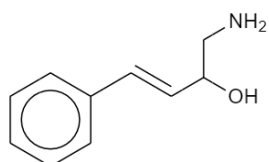
constitutional isomer (structural isomer)

(b) (i) HCN, trace amount of NaCN (or trace amount of NaOH)

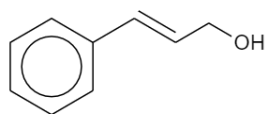
(ii) compound H



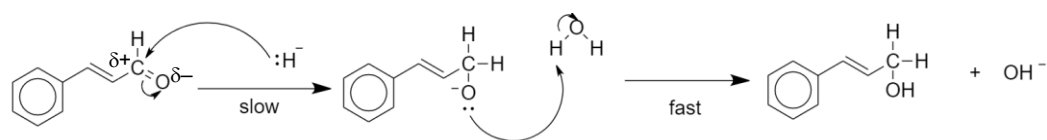
compound I



(c) (i)



(ii)

(Note: H^- is not regenerated as it is not a catalyst)

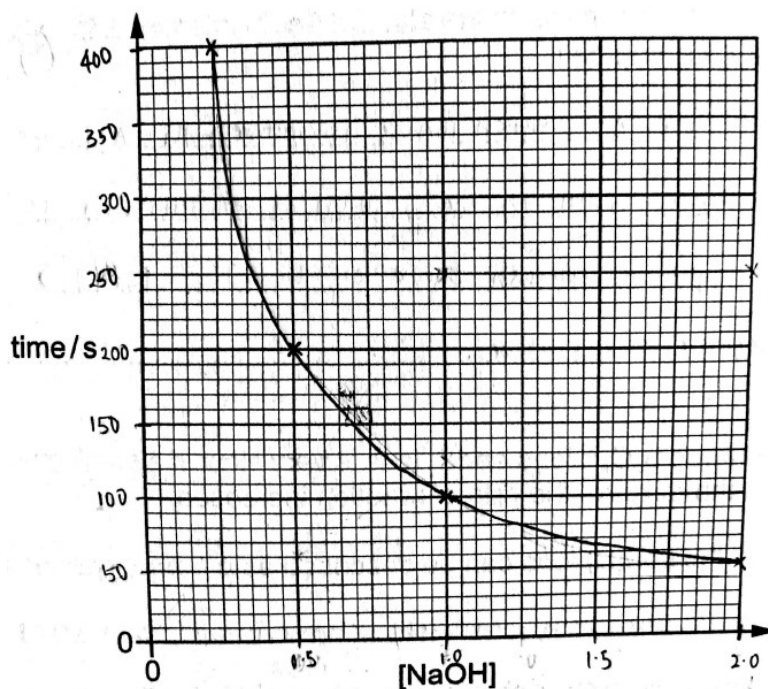
- (iii) As $A/$ is larger than B , the valence orbital overlap less effectively with the orbital of H , making A/H bond weaker than B/H bond. As a result, the A/H bond is broken more easily, and releases the H^- more readily.

OR

$A/$ is more electropositive than B . The A/H bond is more polarised than the B/H bond and is easier to form the H^- nucleophile (by breaking the A/H bond).

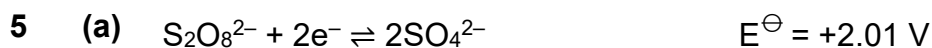
- (d) (i)
1. For experiment 2, to a 20 cm^3 portion of cinnamaldehyde, add 10 cm^3 of 2 mol dm^{-3} NaOH and 2 cm^3 of deionised water. Mix thoroughly until a clear mixture is produced. Add 15 drops of propanone with continuous stirring, and use a stopwatch to record the time taken (to the nearest second) for a precipitate to appear.
 2. Repeat step 1 for experiments 3 to 6 with 20 cm^3 portions of cinnamaldehyde with 8 cm^3 , 6 cm^3 , 4 cm^3 , 2 cm^3 of 2 mol dm^{-3} NaOH and 4 cm^3 , 6 cm^3 , 8 cm^3 , 10 cm^3 of deionised water respectively.
 3. Since the total volume is kept constant, volume of NaOH will be directly proportional to $[\text{NaOH}]$. Plot a graph of $\frac{1}{\text{time}} / \text{s}^{-1}$ against volume of $\text{NaOH} / \text{cm}^3$ and note the shape of the graph.

(ii)



Since rate is proportional to $[\text{NaOH}]$ and the time taken for the precipitate to appear is 50 s for 2 mol dm^{-3} NaOH , rate is halved (time is doubled) when the concentration of NaOH is halved.

(Note: despite some answers correctly describing the quantitative relationship between $[\text{NaOH}]$ and time taken for ppt to appear, many candidates struggled to express this quantitatively in the graph)



$$E^\ominus_{\text{cell}} = +2.01 - (+0.54) = +1.47 \text{ V}$$

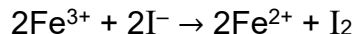
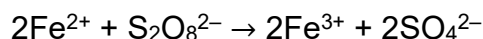
$$\Delta G^\ominus = -(2)(96500)(+1.47) = -283710 \text{ J mol}^{-1} = \underline{-284 \text{ kJ mol}^{-1}}$$

Since $\Delta G^\ominus$ is negative, the reaction is spontaneous.

- (b) AgNO_3 dissolves to give $\text{Ag}^+(\text{aq})$, which forms a (yellow) precipitate, AgI . The removal of I^- from the aqueous solution and the lowering of $[\text{I}^-]$ results in the position of equilibrium of $\text{I}_2 + 2\text{e}^- \rightleftharpoons 2\text{I}^-$ shifting to the right to replenish some of the lost I^- . $E(\text{I}_2/\text{I}^-)$ becomes more positive, hence E_{cell} becomes less positive.

- (c) (i) $\text{S}_2\text{O}_8^{2-}$ and I^- are both negatively charged ions that repel each other, resulting in the reaction having a very high activation energy (E_a). $\text{Fe}^{2+}(\text{aq})$ is in the same phase as $\text{S}_2\text{O}_8^{2-}(\text{aq})$ and $\text{I}^-(\text{aq})$, providing an alternative pathway with a lower E_a to speed up the reaction. $\text{Fe}^{2+}(\text{aq})$ is not consumed/regenerated at the end of the reaction.

- (ii) Iron can form ions of variable oxidation states due to the close proximity in energy of the 3d and 4s electrons, allowing easy conversion from one oxidation state to another.



- (d) In the presence of ligands, the splitting of degenerate d-orbitals in the transition metal ion affects the energy gap (ΔE) between the two sets of d-orbitals. The colour observed is dependent on ΔE .

Different ligands associated with the same ion causes ΔE to be different. The different OH^- ligand causes ΔE to be different so the colour observed is different

(Note: ΔE for $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ is smaller than that of $[\text{Fe}(\text{H}_2\text{O})_5(\text{OH})]^{2+}$, resulting in $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ absorbing yellow light which is lower energy than the violet light absorbed by $[\text{Fe}(\text{H}_2\text{O})_5(\text{OH})]^{2+}$).

The different electronic configuration due to difference in oxidation state of the metal also affects ΔE so colour observed is different even though the ligands are the same.

(Note: ΔE for $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ is smaller than that of $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$, resulting in $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ absorbing red light which is lower energy than the yellow light absorbed by $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$).

- 6 (a) (i) Benzoic acid is stable / benzoic acid is easy to obtain in the pure form / benzoic acid is stable and does not decompose on its own / benzoic acid is a solid that does not vaporise easily & makes it easy to handle.
- (ii) The temperature of the water jacket can be controlled to be similar to that inside the calorimeter such that the temperature gradient inside and outside the calorimeter is negligible to minimise heat loss.
- (b) (i) The black powder is carbon soot, suggesting there was incomplete combustion of benzoic acid/not enough oxygen. The amount of energy released was less than expected, hence the water was heated up to a lower than expected temperature.
- (ii) Use an excess of oxygen gas / increase the pressure of oxygen gas / flush the crucible a few times with oxygen.
- (iii) Average ΔT from samples 2 & 3 = $\frac{(55.3-25.0)+(54.7-25.0)}{2} = 30\text{ }^{\circ}\text{C}$
- No. of moles of benzoic acid = $\frac{6.10}{122} = 0.0500\text{ mol}$
- Heat evolved = $(0.0500)(3230) = 161.5\text{ kJ} = 161500\text{ J}$
- $161500 = \text{heat capacity} \times 30$
- Heat capacity = $5383\text{ J }^{\circ}\text{C}^{-1} = \underline{5.38\text{ kJ }^{\circ}\text{C}^{-1}}$
- (c) (i) $\text{C}_6\text{H}_5\text{COOH(s)} + \frac{15}{2}\text{O}_2\text{(g)} \rightarrow 7\text{CO}_2\text{(g)} + 3\text{H}_2\text{O(l)}$
- (Note: many candidates did not include state symbols or showed water as a gas; the use of 1 mole of benzoic acid is crucial as this is a question on standard enthalpy change of combustion)
- (ii) Determination of standard enthalpy change of combustion will need to be done under standard conditions (298K and 1 bar). However, conditions in the calorimeter are not under standard conditions (high pressure and temperature).

7 (a) (i) $\Delta H = -118 \times 5 = \underline{-590 \text{ kJ mol}^{-1}}$

(ii) The actual value is less exothermic than that in (a)(i), suggesting that the actual molecule is more stable than the hypothesised one with 5 C=C bonds. Naphthalene is energetically more stable.

(iii) Each C atom is sp² hybridised. The sp² hybrid orbital of one C atom overlaps head-on with the sp² hybrid orbital of its adjacent carbon atoms, forming σ bonds.

The unhybridised p orbital of each of the 10 C atoms is perpendicular to the ring of C atoms, and they overlap sideways to form an extensive/continuous delocalised π electron system.

(Note: many answers did not make any reference to the bonds between C atoms despite having descriptions of sp² hybridisation)

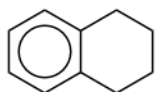
(b) (i) No. of moles of naphthalene = $\frac{0.32}{128} = 0.00250 \text{ mol}$

No. of moles of hydrogen gas = $\frac{(101000)(125 \times 10^{-6})}{(8.31)(30+273)} = 0.00501 \text{ mol}$

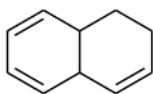
$\frac{n_{H_2}}{n_{\text{naphthalene}}} = \frac{0.00501}{0.00250} = 2.01 \approx 2$

1 molecule of naphthalene (C₁₀H₈) reacts with 2 molecules of H₂ during hydrogenation to form the product. Hence, the product will have 4 more H atoms than naphthalene, resulting in its formula to be C₁₀H₁₂.

(ii) Compound **K**



Compound **L**



(iii) Compound **K** is more likely to be formed. The resonance-stabilised aromatic structure is unlikely to undergo addition of hydrogen as destroying the resonance structure to form compound **L** is energetically not feasible.

(iv) H₂(g), Pt/Ni catalyst, heat at high temperature and pressure

Suggested solutions for Paper 3

- 1 (a) HCl shows little tendency to decompose. HBr produces reddish brown bromine vapour on strong heating. HI gives copious violet fumes of iodine when a red-hot needle is plunged into a jar containing hydrogen iodide.

The size of the halogen atoms increases from Cl to Br to I and their valence orbitals become more diffuse. This results in less effective overlap of the orbitals between the small H atom and the larger halogen atom and less energy is required to break the weaker H-X bond, resulting in a decrease in thermal stability from HCl to HBr to HI .

(Note: Many candidates only explained the trend in the thermal stabilities, without initially describing the reaction.)

Misconception: Several candidates stated that the variation in bond length was responsible for the trend, without relating this to bond strength.)

- (b) (i) MgCl_2 has a giant ionic lattice structure while SiCl_4 has a simple covalent structure. More energy is needed to overcome the stronger electrostatic attraction between Mg^{2+} and Cl^- ions in MgCl_2 than the weak instantaneous dipole-induced dipole interactions (id-id) between SiCl_4 molecules. Hence MgCl_2 has a higher melting point than SiCl_4 .

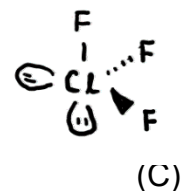
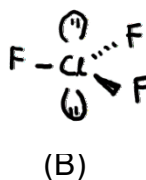
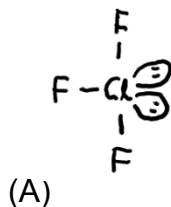
(Misconceptions: Some candidates described the intermolecular forces in SiCl_4 as being due to weak covalent bonds and some described the intermolecular forces as being due to permanent dipoles.)

- (ii) Both SiCl_4 and SiF_4 have simple covalent structure. SiCl_4 has more electrons than SiF_4 . More energy is needed to overcome the stronger id-id interactions between SiCl_4 molecules than that between SiF_4 molecules. Hence SiCl_4 has a higher melting point than SiF_4 .

(Note: Some incorrectly used arguments based on electronegativities of the elements.)

- (iii) Both MgF_2 and MgCl_2 have giant ionic lattice structure. F^- is smaller than Cl^- . More energy is needed to overcome the stronger electrostatic attraction between Mg^{2+} and F^- ions than that between Mg^{2+} and Cl^- ions. Hence MgF_2 has a higher melting point than MgCl_2 .

- (c) (i)



no dipole moment

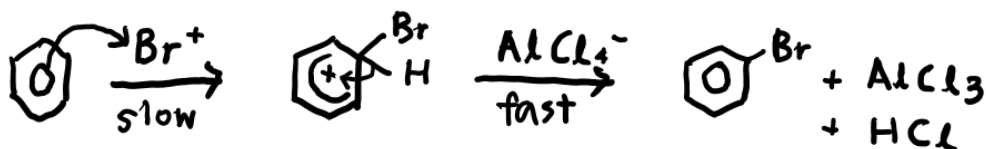
- (ii) (A) is the most stable arrangement as the lone pairs of electrons are placed in the equatorial plane, minimising lone pair-lone pair repulsion which is the strongest form of electronic repulsion in the VSEPR Theory.

Although the lone pair-lone pair repulsion is the minimum in (B), there is greater lone pair-bond pair repulsion compared to (A), resulting in intermediate stability of (B).

The lone pair-lone pair repulsion and lone pair-bond pair repulsions in (C) are both strong, hence (C) is the least stable arrangement.

- (d) (i) Product is bromobenzene.

Br is less electronegative than Cl, hence the electrophile generated from the reaction between BrCl and AlCl₃ is Br⁺.



(Note: The mechanism was generally well described, though some candidates did not use AlCl₃ in the production of the electrophile. The placement of the '+' charge in the intermediates, were sometimes inexact.)

- (e) (i) $K_{sp} = [\text{Ag}^+][\text{Cl}^-]$

Units: mol² dm⁻⁶

(ii)

	AgCl(s)	⇌	Ag ⁺ (aq)	+	Cl ⁻ (aq)
Initial [] / mol dm ⁻³			0.50		0
Change in [] / mol dm ⁻³			+ x		+ x
Eqm [] / mol dm ⁻³			0.50 + x		x

$$2.0 \times 10^{-10} = (0.50 + x)(x)$$

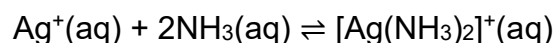
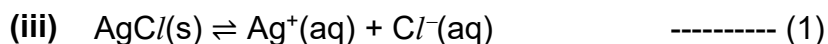
Assuming that x is small such that $0.50 + x \approx 0.50$

$$2.0 \times 10^{-10} = (0.50)(x)$$

$$x = 4.00 \times 10^{-10}$$

$$[\text{Cl}^-] = \underline{4.00 \times 10^{-10} \text{ mol dm}^{-3}}$$

(Note: Many candidates obtained the correct answer, but some did not take into account the 0.5 mol dm⁻³ [Ag⁺(aq)] and took the square root of K_{sp}.)

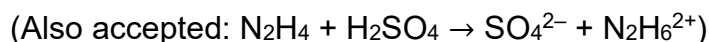
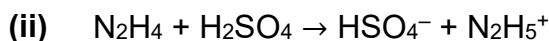


$\text{NH}_3(\text{aq})$ forms a soluble complex $[\text{Ag}(\text{NH}_3)_2]^+(\text{aq})$ with $\text{Ag}^+(\text{aq})$. This causes the equilibrium position of (1) to shift right to replenish some of the $\text{Ag}^+(\text{aq})$ consumed, resulting in more AgCl(s) dissolving. Hence, solubility of AgCl(s) increases.

NaCl is completely soluble, increasing the $[\text{Cl}^-(\text{aq})]$. Due to the common ion effect, this causes the equilibrium position of (1) to shift left to consume some of the added $\text{Cl}^-(\text{aq})$, producing more AgCl(s) . Hence, solubility of AgCl(s) decreases.

- 2 (a) (i) The hydrogen bonds between hydrazine molecules are more extensive than that between ammonia molecules due to the additional lone pair of electrons on the extra N atom. Hydrazine forms an average of 2 hydrogen bonds per molecule whereas ammonia forms an average of 1 hydrogen bond per molecule. More energy is required to overcome the hydrogen bonds in hydrazine.

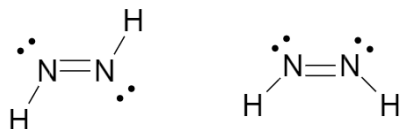
(Misconception: Some candidates thought that hydrazine's higher boiling point was solely due to the extra electrons the molecule contains.)



N_2H_4 is the weak base, N_2H_5^+ is its strong conjugate acid.

H_2SO_4 is the strong acid, HSO_4^- is the weak conjugate base.

(b)



Bond angle = 118° (any value between 114° to 119°)

Cis and trans isomerism arises due to the restricted rotation about the N=N bond.

(c) Gaseous mixture: $\text{NH}_3(\text{g})$, $\text{HN}_3(\text{g})$

Solution of **A** in water: $\text{NH}_4^+(\text{aq})$, $\text{N}_3^-(\text{aq})$



(ii) No. of moles of $\text{NaN}_3 = \frac{400}{65} = 6.15 \text{ mol}$

No. of moles of $\text{N}_2 = \frac{8}{5} \times 6.15 = 9.85 \text{ mol}$

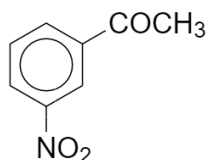
$pV = nRT$

$p(100 \times 10^{-3}) = (9.85)(8.31)(298)$

$p = \underline{2.44 \times 10^5 \text{ Pa}}$

(e) (i) Secondary amide

(ii)

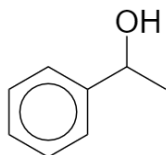


Concentrated HNO_3 , concentrated H_2SO_4 , maintain at $>55^\circ\text{C}$

(iii) Step 3: $\text{CH}_3\text{CH}_2\text{Cl}$, anhydrous AlCl_3

Step 4: limited $\text{Cl}_2(\text{g})$, UV light

(iv) $\text{NaOH}(\text{aq})$, heat under reflux



(v) $\text{NaOH}(\text{aq})$, heat under reflux

(vi) **D** is less basic than **F**. The lone pair of electrons on N atom of the amide group in **D** is delocalised over to the adjacent $\text{C}=\text{O}$ group, making it less available for donation/protonation. Hence, the amide in **D** is neutral / less basic than the phenylamine in **F**.

- 3 (a) Copper has a higher melting point than calcium. Due to the close proximity of the 3d and 4s electrons in copper, they can be contributed to the sea of delocalised valence electrons whereas calcium only contributes the two 4s electrons to the sea of delocalised valence electrons. The resultant copper ions are smaller and the electrostatic attraction between copper ions and its sea of delocalised valence electrons is stronger than that between calcium ions and its sea of delocalised valence electron, requiring more energy to overcome.

Copper has higher density than calcium. Copper has a larger mass and smaller radius than calcium, hence copper has a higher mass per unit volume.

- (b) (i) The electronic configuration of Cu^{2+} is $1s^2 2s^2 2p^6 3s^2 3p^6 3d^9$, hence Cu^{2+} has a partially-filled 3d-subshell. In the presence of ligands, the degenerate 3d orbitals split into two sets with different energies. The energy gap (ΔE) is small. An electron in the lower energy d-orbital is promoted to a partially-filled d-orbital of higher energy when radiation from the visible region of the electromagnetic spectrum is absorbed. The observed colour (blue) is complement to the colour absorbed (orange).

The electronic configuration of Zn^{2+} is $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10}$. Since the d-orbitals are fully-filled, d-d electronic transition cannot occur.

(Misconception: Some candidates suggested either that there was no splitting of the d-orbitals in $\text{Zn}^{2+}(\text{aq})$, or that the splitting was of such a magnitude as not to correspond to the visible region.)

- (ii) $[\text{Cu}(\text{H}_2\text{O})_6]^{2+} + 4\text{NH}_3 \rightleftharpoons [\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+} + 4\text{H}_2\text{O}$

Ligands are ions or atoms or molecules with at least a lone pair of electrons that can be donated into a low-lying vacant/partially-filled orbital of the central metal atom or ion, forming a dative covalent bond. Examples include H_2O in $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ and H_2O & NH_3 in $[\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$ complex.

A complex is a chemical species formed when a central metal atom or ion is bonded with one or more ligands via dative covalent bond. Examples include $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$.

- (c) V_{KMnO_4} needed to oxidise vanadium in **H** to $\text{VO}_2^+ = 24.6 \text{ cm}^3$

V_{KMnO_4} needed to oxidise vanadium in **G** to $\text{VO}_2^+ = 16.4 \text{ cm}^3$

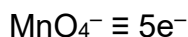
Ratio of titre volume for **H** to VO_2^+ : titre volume for **G** to VO_2^+

= ratio of amount of electrons transferred from **H**: amount of electrons transferred from **G**

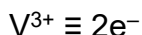
= 3:2

Since the final oxidation state VO_2^+ is +5, oxidation state of vanadium in **H** is +2 (in the form of V^{2+}) [3 mole of electrons transferred to form VO_2^+], and oxidation state of vanadium in **G** is +3 (in the form of V^{3+}) [2 moles of electrons transferred to form VO_2^+].

No. of moles of MnO_4^- for 1st sample = $\frac{16.4}{1000} \times 2.00 \times 10^{-3} = 3.28 \times 10^{-5} \text{ mol}$



No. moles of e^- taken in by $\text{MnO}_4^- = 3.28 \times 10^{-5} \times 5 = 1.64 \times 10^{-4} \text{ mol}$



No. of moles of $\text{V}^{3+} = \frac{1}{2} \times 1.64 \times 10^{-4} = 8.20 \times 10^{-5} \text{ mol}$

$$[\text{V}^{3+}] = 8.20 \times 10^{-5} \div \frac{25.0}{1000} = \underline{3.28 \times 10^{-3} \text{ mol dm}^{-3}}$$

(Note: Since the end product is VO_2^+ where vanadium has an oxidation state of +5, possible oxidation numbers of the reduced vanadium are 0, +2, +3 and +4 (from Data Booklet). Since the vanadium compound in **G** can be reduced by Zn to form a soluble vanadium compound in **H**, **G** would not contain V^{2+} as the product of such a reduction would have been V, which is an insoluble solid metal. Therefore, the oxidation state of vanadium in **G** would have been either +3 or +4.)

- (d) **I** does not undergo acid-metal reaction $\Rightarrow$ **I** is not a phenol or alcohol or carboxylic acid.

Since **I** is not a phenol, it does not undergo electrophilic substitution.

I undergoes electrophilic addition $\Rightarrow$ **I** is an alkene.

The 2 oxygen atoms in **I** are either part of aldehyde or ketone or ester functional groups.

I does not give positive iodoform test, and it is not an alcohol, hence **I** does not have $-\text{COCH}_3$ or $-\text{CH}(\text{OH})\text{CH}_3$ groups.

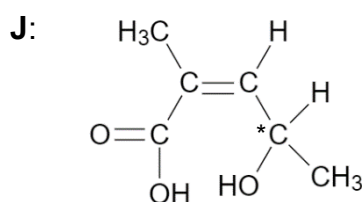
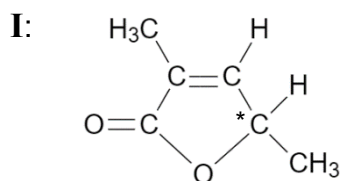
I contains a chiral carbon.

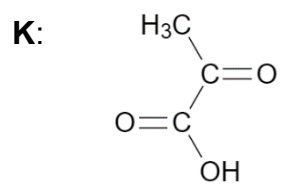
I undergoes acidic hydrolysis to form **J** only $\Rightarrow$ **I** is a cyclic ester and **J** has both alcohol and carboxylic acid groups.

Since **I** has a chiral carbon, the alcohol from the hydrolysis of ester **J** is a secondary alcohol.

J undergoes vigorous oxidation to form one product **K**, which has half the number of carbon atoms of **J** $\Rightarrow$ $\text{C}=\text{C}$ bond cleaved to give 2 molecules of **K**.

K undergoes acid-metal reaction and gives a positive iodoform test $\Rightarrow$ **K** is a carboxylic acid and has $-\text{COCH}_3$ group.





- 4 (a) (i) Let the oxidation number of carbon be x.

$$4x + (+1)(8) + (-2)(2) = 0$$

$$x = -1$$

or

Oxidation number of carbon-1 = +3

Oxidation number of carbon-2 = -2

Oxidation number of carbon-3 = -2

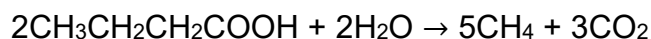
Oxidation number of carbon-4 = -3

$$\text{Average oxidation number} = [+3 + (-2) + (-2) + (-3)] \div 4 = -1$$

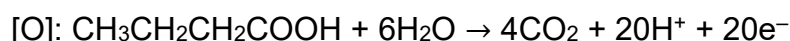
- (ii) Reduction of butanoic acid to CH₄: oxidation state of carbon decreased by 3 units from -1 in butanoic acid to -4 in CH₄.

Oxidation of butanoic acid to CO₂: oxidation state of carbon increased by 5 units from -1 in butanoic acid to +4 in CO₂.

Hence, 5 mol of CH₄ will be formed together with 3 mol of CO₂.



OR



[R] $\times$ 5, [O] $\times$ 3, and simplifying:



- (iii) Pass the gaseous mixture through NaOH(aq) (or any other alkali).

- (iv) Mole ratio of methane in the mixture = $\frac{5}{8}$

$$\text{Pressure of methane} = \frac{5}{8} \times 1.5 \times 10^5 = \underline{9.38 \times 10^4 \text{ Pa}}$$

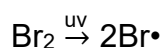
- (b) (i) $\Delta H_r^\ominus = [(3)(-394) + (5)(-75)] - [(2)(-534) + (2)(-286)] = \underline{+83.0 \text{ kJ mol}^{-1}}$

- (ii) $-207 = +83 - (298)(\Delta S_r^\ominus)$

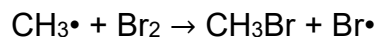
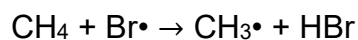
$$\Delta S_r^\ominus = \underline{+0.973 \text{ kJ mol}^{-1} \text{ K}^{-1}}$$

The sign is positive, indicating there is an increase in disorder due to an increase in the number of gaseous molecules from 0 to 8.

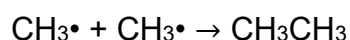
(c) (i) Initiation:



Propagation:



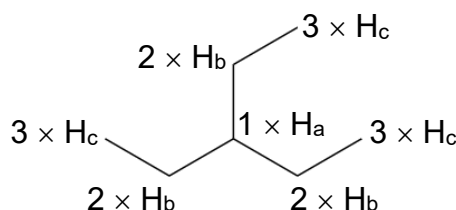
Termination: (write any 2)



(Note: Candidates who performed less well did not identify the correct step that requires light)

(ii) The X–X bond in the halogen molecule is broken homolytically in the initiation step. Br–Br bond is weaker than C–C bond due to the larger size of Br atoms resulting in less effective orbitals overlap. Light of lower energy, and hence longer wavelength, can be used to break the Br–Br bond.

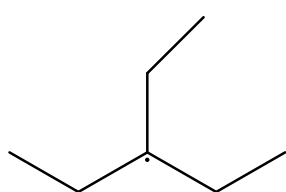
(iii)



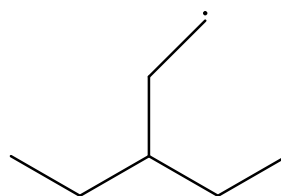
There are three types of hydrogen atoms in 3-ethylpentane ($\text{H}_a/\text{H}_b/\text{H}_c$).

Substitution of H_a gives **L**, H_b gives **M** and H_c gives **N**. Since there are a total of 1 H_a , 6 H_b and 9 H_c , the ratio of **L** : **M** : **N** will be 1 : 6 : 9.

(iv) Instead of the predicted 1 : 9 ratio, the 2 : 1 ratio observed suggests that **L** is 18 times more likely to be formed compared to **N**. The intermediate to form **L** is a tertiary radical that is stabilised by 3 electron donating alkyl groups, while the intermediate to form **N** is a primary radical that is less stable. Hence the observed ratio favours the formation of **L** due to the more stable intermediate.



Intermediate to form **L**



Intermediate to form **N**

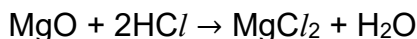
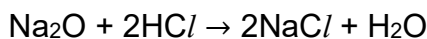
(d) **Q:** $\text{CH}_3\text{CH}_2\text{CHC}/\text{COO}^-\text{Na}^+$

R: $\text{CH}_3\text{CH}_2\text{CH}(\text{NHCH}_2\text{CH}_3)\text{COO}^-\text{Na}^+$

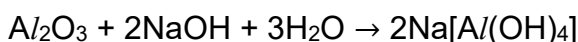
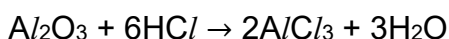


- 5 (a)** Across the period, oxides vary from ionic to covalent.

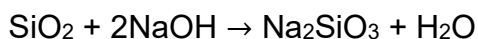
Na_2O and MgO are ionic oxides that are basic and react with acids to form salt and water.



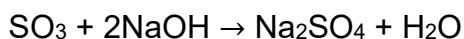
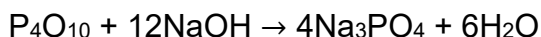
Al_2O_3 , an ionic oxide with covalent character, is amphoteric and reacts with both acids and alkalis to form salt (and water for the reaction with acid).



SiO_2 is a covalent oxide that is acidic and reacts with concentrated alkalis to form salt and water.

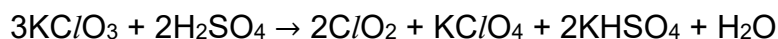


P_4O_{10} and SO_3 are covalent oxides that are acidic and react with alkalis to form salt and water.

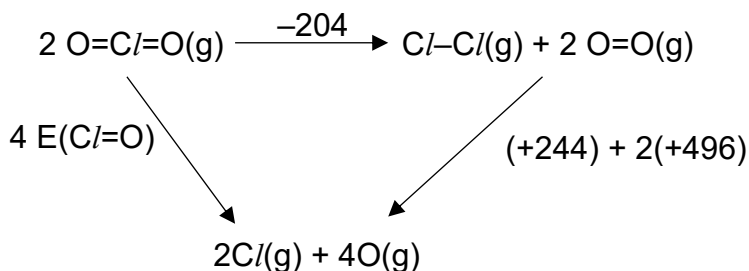


- (b) (i)** Oxidation number of chlorine decreases by 1 unit from +5 in KClO_3 to +4 in ClO_2 and increases by 2 units from +5 in KClO_3 to +7 in KClO_4 .

Hence, 2 mol of ClO_2 will be formed with 1 mol of KClO_4 .



(ii)



$$4 \text{ E}(\text{Cl}=\text{O}) = -204 + (+244) + 2(+496)$$

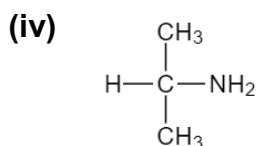
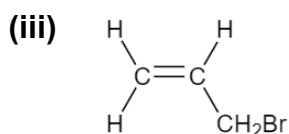
$$\text{E}(\text{Cl}=\text{O}) = \underline{+258 \text{ kJ mol}^{-1}}$$

(Note: The most common error was forgetting that the ClO_2 molecule contains two $\text{Cl}=\text{O}$ bonds.)

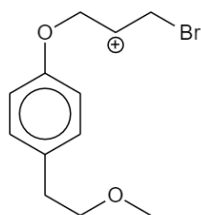
- (iii)** As the number of gaseous particles increases from 2 to 3, $\Delta S^\ominus$ will be positive and $-\text{T}\Delta S^\ominus$ will be negative.

$\Delta G^\ominus$ also has to be negative as decomposition of C/O_2 is spontaneous. Since $\Delta G^\ominus = \Delta H^\ominus + (-T\Delta S^\ominus)$, and both $\Delta H^\ominus$ and $-T\Delta S^\ominus$ are both negative, $\Delta G^\ominus$ will be more negative than $\Delta H^\ominus$.

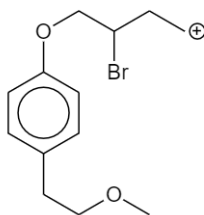
- (c) (i) Step 3: reduction
Step 4: acid-base reaction
- (ii) Step 1: electrophilic substitution
Step 2: $\text{S}_\text{N}2$ nucleophilic substitution
Step 6: electrophilic addition



- (v) After the electrophilic attack by the $\text{C}=\text{C}$ bond on the $\text{Br}^{\delta+}$ electrophile, the



more stable secondary carbocation will be formed as there is 1 more electron-donating alkyl group to stabilise the positive charge. This carbocation will be attacked by H_2O to give the isomer in Figure 5.1.



The less stable primary carbocation, which will give the isomer in Figure 5.2, will not be formed.